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Shaw et al.

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[54] **FIBER OPTIC AMPLIFIER**

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[73] Assignee: **The Board of Trustees of the Leland Stanford Junior University**, Stanford, Calif.

[*] Notice: The portion of the term of this patent subsequent to Nov. 19, 2002 has been disclaimed.

[21] Appl. No.: **137,870**

[22] Filed: **Dec. 22, 1987**

Related U.S. Application Data

[62] Division of Ser. No. 930,136, Nov. 12, 1986, Pat. No. 4,723,824, which is a division of Ser. No. 554,888, Nov. 25, 1983, Pat. No. 4,674,830.

[51] **Int. Cl.**[4] **G02B 6/26; H01S 3/07**

[52] **U.S. Cl.** **350/96.15;** 330/4.3; 372/6; 372/69

[58] **Field of Search** 350/96.15, 96.16; 372/6, 69, 94; 330/4, 4.3, 5, 84, 85

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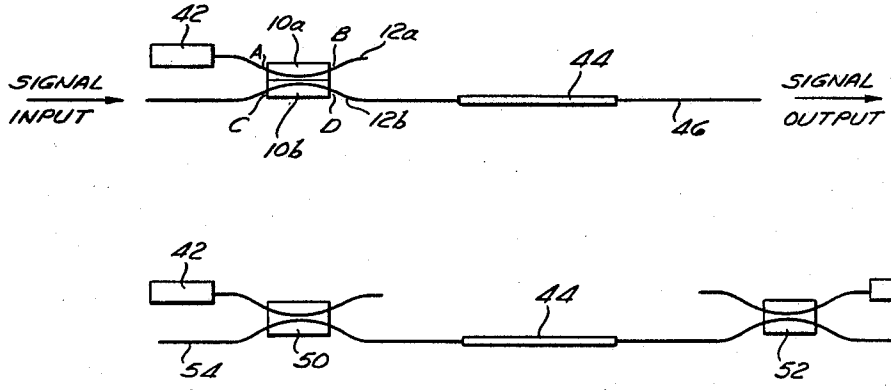
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[57] **ABSTRACT**

An amplifier for use with fiber optic systems comprises a neodymium YAG crystal placed in series with a signal-carrying optical fiber. The ND:YAG crystal is supplied by the optical fiber with both the signal to be amplified, and pumping illumination. The pumping illumination is coupled onto the optical fiber by a multiplexing coupler which is used to combine the signal to be amplified and illumination from a pumping illumination source onto a single optical fiber. The pumping illumination inverts the neodymium ions within the ND:YAG crystal. The signal to be amplified propagates through this crystal to stimulate emission of coherent light from the neodymium ions, resulting in amplification of the signal. Because this arrangement permits the ND:YAG crystal to be end-pumped with pumping illumination, and because the length of the ND:YAG crystal may be substantially greater that the absorption length for the crystal at the wavelength of the pumping illumination, virtually all of the pumping illumination may be absorbed within the ND:YAG crystal and used for amplification of the signal carried by the optical fiber.

15 Claims, 5 Drawing Sheets



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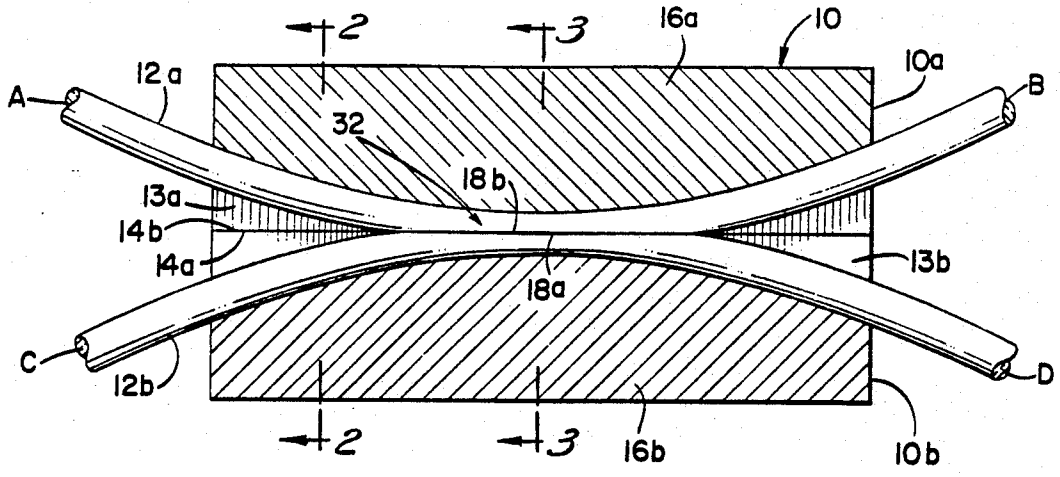


Fig. 1

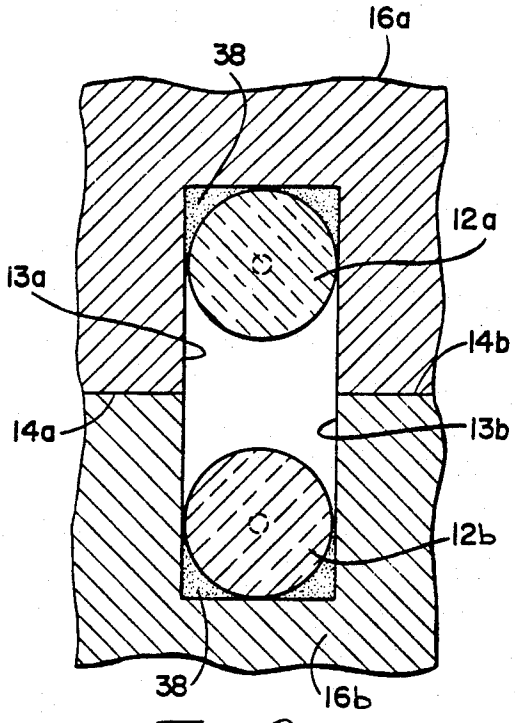
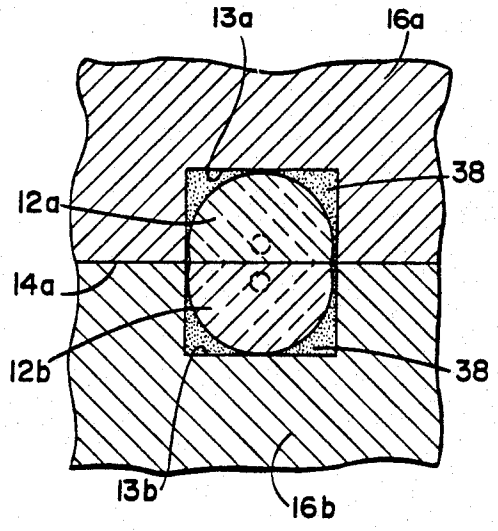


Fig. 2

Fig. 3



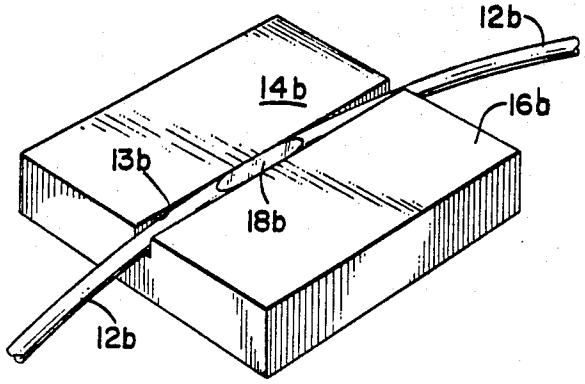


Fig. 4

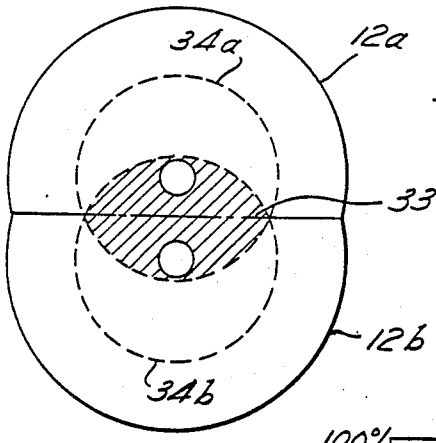


Fig. 5

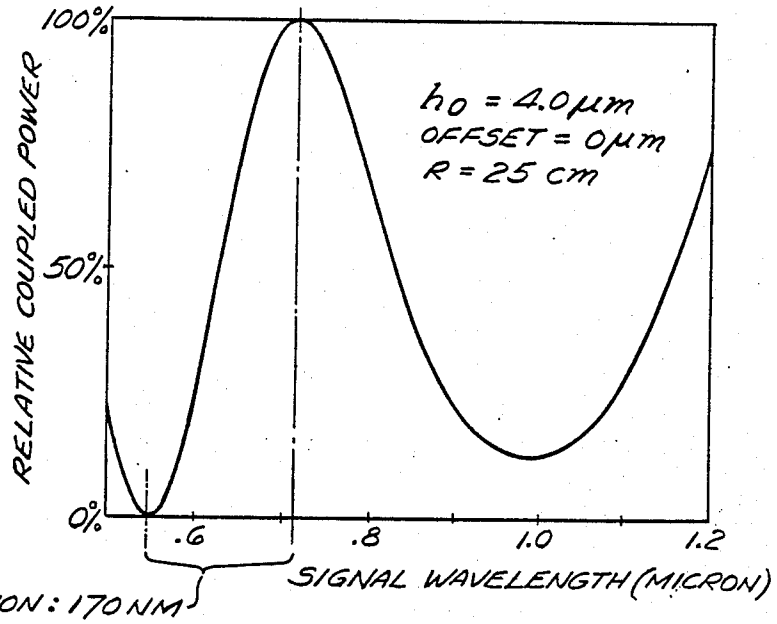


Fig. 6

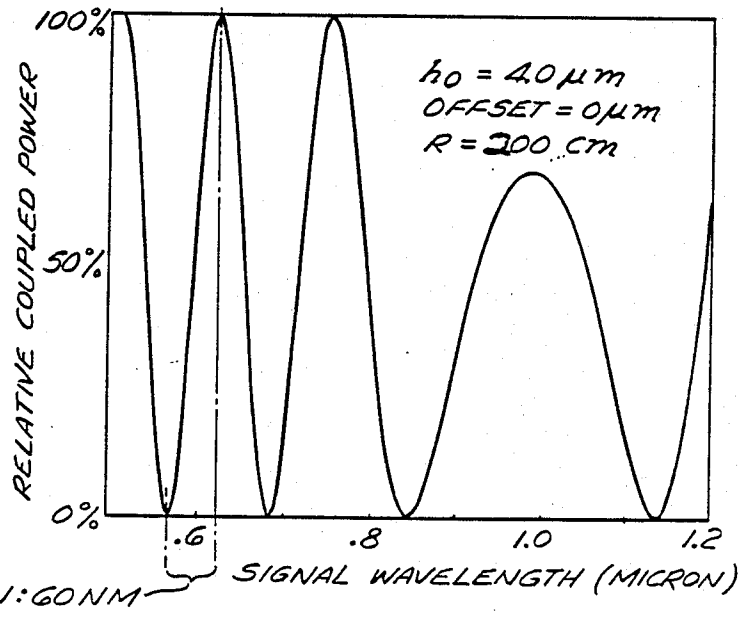


Fig. 7

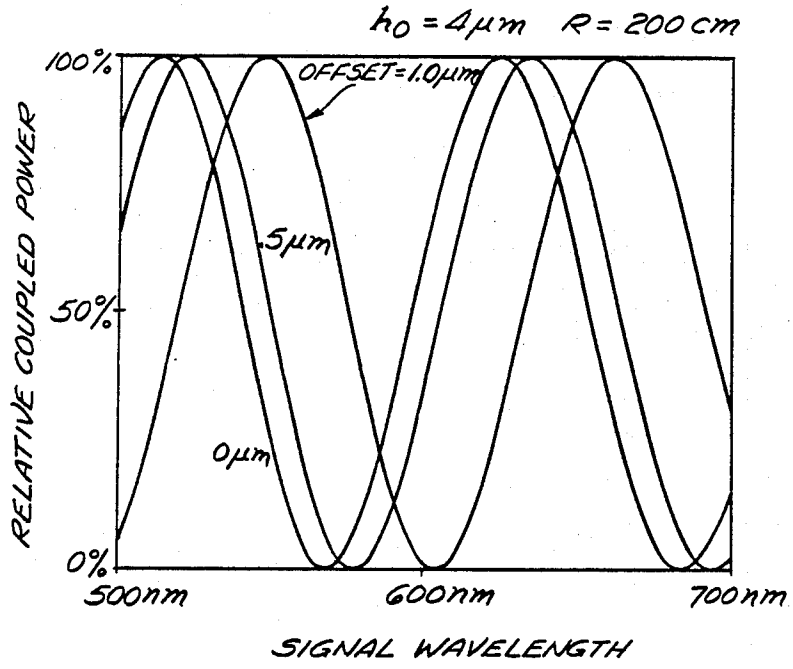

$h_0 = 4\mu m$ $R = 200\,cm$
100%
50%
0%
OFFSET=1.0μm
5μm
0μm
RELATIVE COUPLED POWER
500nm
600nm
700nm
SIGNAL WAVELENGTH

Fig. 8

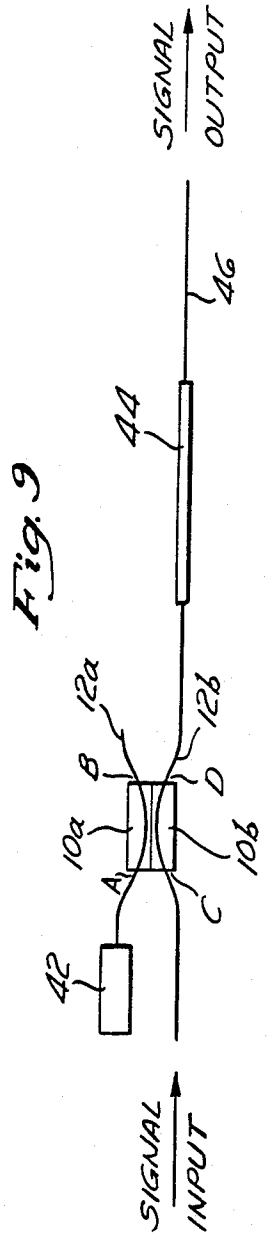


Fig. 9

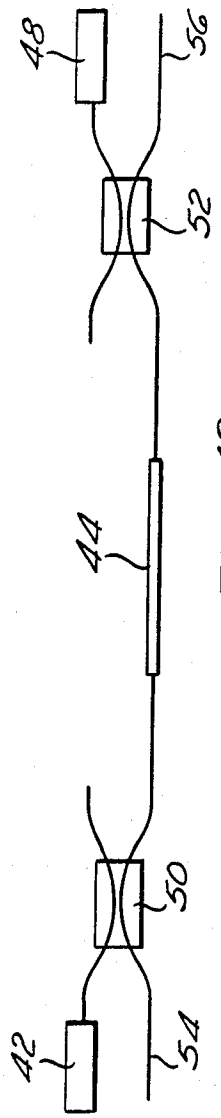


Fig. 12

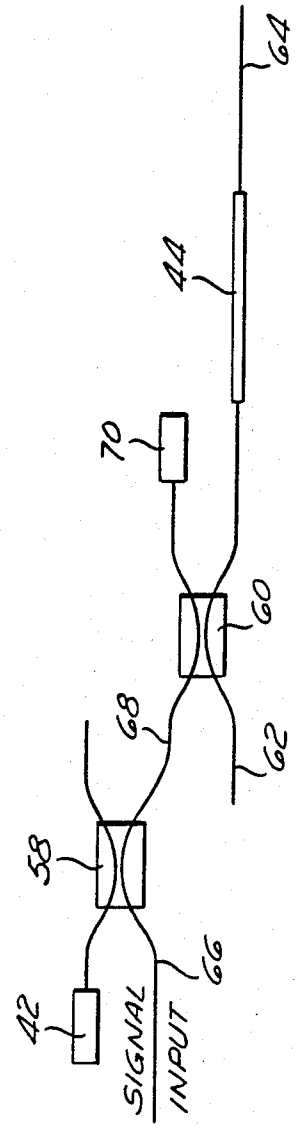


Fig. 13

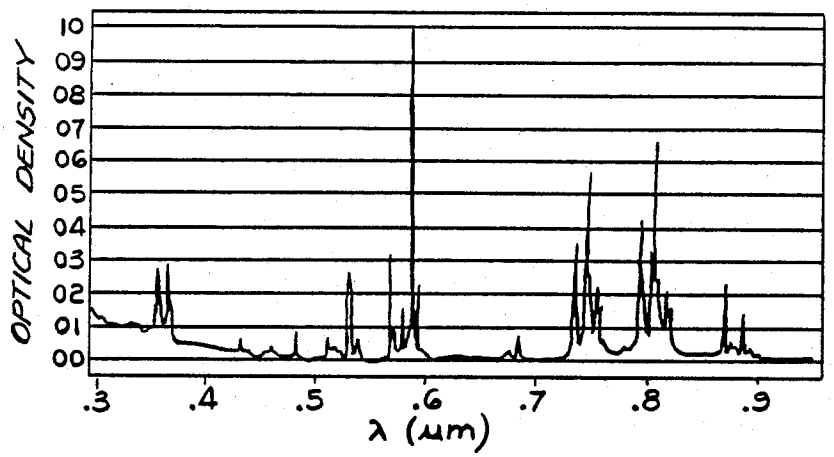


Fig. 10

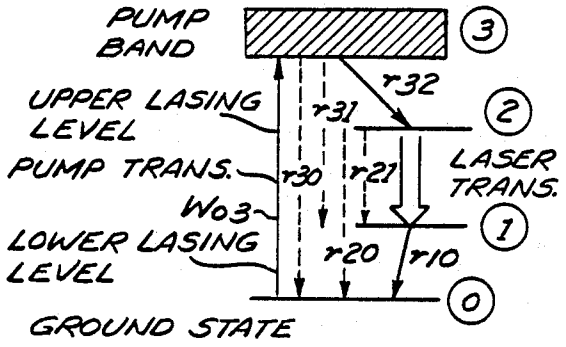


Fig. 11a

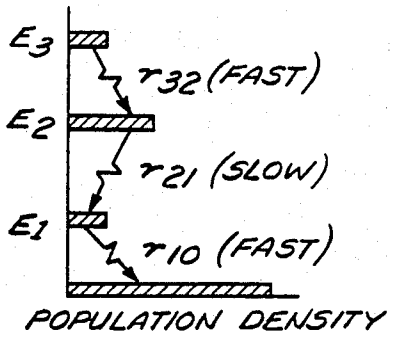


Fig. 11b

FIBER OPTIC AMPLIFIER

This application is a division of application Ser. No. 930,136, filed Nov. 12, 1986 U.S. Pat. No. 4,723,824, which is a division of application Ser. No. 554,888 filed Nov. 25, 1983, now issued as U.S. Pat. No. 4,674,830.

BACKGROUND OF THE INVENTION

The concept of optical amplifiers, based upon the capability of certain materials, particularly on a macroscopic level, is well known. Thus, for example, it is known to place a pumping light source and a single crystal neodymium-yttrium aluminum garnet (ND:YAG) rod, several millimeters in diameter and several centimeters in length, in a tubular reflective cavity. For example, the light source and ND:YAG rod may be located, respectively, to extend along the two foci of a cavity having an elliptical cross section. In such an arrangement, light emitted by the light source and reflected from the cavity walls will impinge upon the ND:YAG rod. The light source is preferably selected to emit wavelengths corresponding to the absorption spectra of the ND:YAG crystal so that the neodymium ions of the crystal are inverted to an energy level above the upper lasing level. After inversion, an initial relaxation of the neodymium ions through phonon radiation yields an ion population at the upper lasing level. From this level, the ions will relax to a lower lasing level, emitting light of a wavelength which is characteristic of the ND:YAG material. Advantageously, the lower lasing level is above the ground level for the ions so that a rapid, phonon-emitting relaxation will occur between the lower lasing level and the ground level, enabling a high inversion ratio to exist between the upper and lower lasing levels within the pumped ions.

With the population so inverted, as is well known from laser technology, the ND:YAG will provide a very slow fluorescence, that is, random emission of incoherent light. This spontaneous radiation, however, has a minimal effect on the amplifying rod, since the average lifetime of ions in the inverted state is 230 microseconds, in ND:YAG.

If, after some of the neodymium ions of the ND:YAG rod have been inverted, a light signal at the lasing frequency is transmitted through the rod, the light signal will trigger the relaxation of the neodymium ions, causing coherent emission of stimulated radiation, which will effectively add to the transmitted signal, thus amplifying this signal.

The absorption length of the pumping illumination within the ND:YAG crystal (i.e., the length of material through which the illumination must traverse before about 65% of the illumination is absorbed) is typically in the range between 2 and 3 millimeters, and thus the ND:YAG crystals used in transverse pumping structures such as described previously have had diameters at least this large so that the crystal could absorb a substantial portion of the pumping radiation during the initial reflection from the cavity walls and passage through he crystal. If, during this initial traverse through the crystal, the pumping illumination is not absorbed, it is likely to be reflected by the cavity walls back to the light source, where it will be reabsorbed, generating heat in the light source and reducing the overall efficiency of the amplifier.

When such amplifiers are used in fiber optic systems, it has been thought necessary, because of the large difference in diameter between the optical fiber and the ND:YAG crystal, to use optical components, such as lenses, to focus light from the optical fiber into the ND:YAG rod, and the amplified light signal from the ND:YAG rod back into another fiber. Such optical systems require careful alignment and are susceptible to environmental changes, such as vibration, and thermal effects. Additionally, the optical components and the size of the ND:YAG rod make the amplifying system relatively large, and thus impractical for certain applications. Furthermore, the relatively large size of the ND:YAG rod introduces beam wander within the rod due to thermal effects. Thus, the signal from the input fiber optic element will traverse different paths through the rod, a characteristic which is temperature related and varies with time, so that the output light may be lost due to the fact that the fiber will accept only light within a small acceptance angle. Thus, as the beam within the ND:YAG rod wanders, the output signal may vary in an uncontrollable manner. Furthermore, the large size of the ND:YAG rod requires a large amount of input energy in order to maintain a high energy density within the rod. Such large pump power requires high output light sources, generating substantial heat which must be dissipated, typically by liquid cooling of the cavity.

While amplifiers of this type are useful in many applications, such as some communications applications, a use which puts severe limitations upon the amplification system is a recirculating fiber optic gyroscope. With such gyroscopes, an optical fiber, typically a kilometer or more in length, is wound into a loop, and a light signal is recirculated within the loop in both directions. Motion of the loop causes a phase difference between the counter-propagating light signals which may be used to measure gyroscope rotation. It is advantageous, because the phase shift induced in one rotation is relatively small and because periodic outputs relating to rotation are required, to recirculate input light within the loop as many times as possible.

In traversing a kilometer of optical fiber, an optical signal will typically lose 30 to 50 percent of its intensity. An amplifier, if capable of amplifying the bidirectional counter-propagating light signals, would permit a light signal to propagate many more times within the loop, if the amplifier were placed in a series with the loop, and provided a gain equal to the propagation loss.

Unfortunately, the relatively large size, high power requirements caused by relatively inefficient performance, beam wander effects, and cooling requirements of prior art ND:YAG rod amplifiers makes such amplifiers relatively impractical for high accuracy gyroscopes. These factors, of course, also limit the utility of such amplifiers in other applications, such as communication networks.

SUMMARY OF THE INVENTION

The disadvantages associated with crystal rod amplifiers are alleviated in the present invention. This invention permits end pumping of the ND:YAG material, and thus, completely avoids the requirement for a large diameter for this crystal which is inherent in side pumping arrangements. The ND:YAG fiber may thus be made extremely small in diameter in comparison with prior art rod amplifiers, since the pumping illumination is absorbed along the length of the fiber, rather than

across its width. This results in a higher concentration of pumping illumination within the small diameter of the ND:YAG crystal and thus a higher potential gain for the amplifying structure.

In order to accomplish this end pumping, the ND:YAG material is formed as a small diameter fiber and is placed in series with the optical fiber which is transmitting the signal to be amplified.

Adjacent to the ND:YAG fiber, the optical fiber passes through a multiplexing optical coupler. Within this multiplexing coupler, a pair of optical fibers are arranged with a carefully selected interaction length to provide a high fiber-to-fiber coupling efficiency at the wavelength of the pumping source, but a low coupling efficiency at the wavelength of the signal to be amplified. This results in a coupling of the pumping illumination into the optical fiber which is carrying the signal, and thus into the ND:YAG fiber, while substantially eliminating loss to the optical signal which is to be amplified, since this signal is not coupled from the optical fiber of the multiplexing coupler.

Because the present invention permits the pumping illumination to be coupled into the optical signal fiber for guiding to the end of the ND:YAG fiber, the diameter of the ND:YAG fiber need not exceed the absorption length at the pumping wavelength, since the pumping illumination is effectively absorbed in a direction along the axis of the ND:YAG fiber rather than perpendicular to that axis.

Using this arrangement, pumping illumination can be continuously supplied to the amplifying ND:YAG fiber without interfering with its signal carrying characteristics. Thus, since a four-port coupler is used for multiplexing the pumping illumination to the amplifying fiber, the ends of the amplifying fiber are always available for direct signal coupling to the optical fibers within the optical system so that careful time sequencing between the application of pumping illumination and the signal to be amplified is not necessary.

In order to achieve uniform bidirectional amplification within the ND:YAG crystal, pumping illumination may be supplied by multiplexing couplers arranged at both ends of the ND:YAG fiber, providing a symmetrical inversion population along the length of this fiber.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other advantages of the present invention are best understood through reference to the drawings, in which:

FIG. **1** is a cross sectional view of the fiber optic coupler used as a multiplexer in the present invention showing a pair of fiber optic strands mounted in respective arcuate grooves with a given radius of curvature of respective bases;

FIGS. **2** and **3** are cross sectional views of the coupler of FIG. **1** taken along lines **2—2** and **3—3**, respectively;

FIG. **4** is a perspective view of the lower base of the coupler of FIG. **1** separated from the other base to show its associated fiber mounting and the oval-shaped facing surface of the fiber;

FIG. **5** is a schematic diagram showing the evanescent fields of the pair of fibers overlapping at the interaction region;

FIG. **6** is a chart showing relative coupled power versus signal wavelength for a fiber coupler having a minimum fiber spacing of 4 microns, an offset of 0 micron, and a fiber radius of curvature of 25 centimeters;

FIG. **7** is a chart of relative coupled power versus signal wavelengths similar to FIG. **6** but with a fiber radius of curvature of 200 centimeters;

FIG. **8** is a chart of relative coupled power versus signal wavelength for a fiber coupler having a minimum fiber spacing of 4 microns, a fiber radius of 200 centimeters, and a selectable fiber offset;

FIG. **9** is a schematic diagram showing the directional amplifier of the present invention;

FIG. **10** is a diagram showing the absorption spectrum of ND:YAG at 300° K.;

FIG. **11***a* and **11***b* are simplified energy level diagrams of a four-level laser using a doped material, such as ND:YAG;

FIG. **12** is a schematic diagram of a symmetrical bidirectional amplifier in accordance with the present invention;

FIG. **13** is a schematic diagram of an amplifier system, including a multiplexing coupler for signal insertion and signal sensing in a closed fiber loop.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

In order to gain a detailed understanding of the operation of the fiber optic amplifier of this invention, it is necessary to understand that manner in which a fiber optic coupler can be constructed to selectively couple a first optical frequency while not coupling a second optical frequency. The apparatus, as well as methods for constructing the apparatus, required for such selective coupling, are disclosed in copending patent application entitled "Passive Fiber Optic Multiplexer", filed in the U.S. Pat. Office Nov. 9, 1981, bearing Ser. No. 319,301, and listing Herbert J. Shaw and Michel J. F. Digonnet as inventors (now U.S. Pat. No. 4,556,279). The application is assigned to the assignee of the present invention. That application is hereby incorporated herein by reference. Nonetheless, the principal characteristics of that device and its method of manufacture are described below.

This invention utilizes a passive multiplexer which utilizes a fiber optic coupler. This coupler **10** is illustrated in FIGS. **1–4**, and includes two strands **12A** and **12B** of a single mode fiber optic material. The fiber strands **12A** and **12B**, in the embodiment shown, have a core radius of **2** microns, a core refractive index of **1.46** and a cladding index of 1.456. The strands **12A** and **12B** are mounted in longitudinal arcuate grooves **13A** and **13B**, respectively, formed in optically flat confronting surfaces **14A** and **14B**, respectively, of rectangular bases or blocks **16A** and **16B**, respectively. The block **16A** with the strand **12A** mounted in the groove **13A** will be referred to as the coupler half **10A** and the block **16B** with the strand **12B** mounted in the groove **13B** will be referred to as the coupler half **10B**.

Each of the strands **12A** and **12B** comprise an optical fiber which is doped to have a central core and an outer cladding. The strands **12A** and **12B** may comprise a commercially available fiber of quartz glass which is doped to have a central core and an outer cladding. The index of refraction of the fibers **12A** and **12B** should be as nearly as possible identical, and both of the strands **12A** and **12B** should include a central core which is sufficiently small to provide single mode transmission at the optical frequencies to be used. Thus, these strands **12A** and **12B** typically have a core diameter on the order of **10** microns or less and a cladding diameter on the order of **125** microns. In the embodiment disclosed,

the diameter of the strands **12** and their respective cores are exaggerated. As will be understood in more detail from the description which follows, the fiber **12B** is used to transmit the signal to be amplified while the fiber **12A** is used to couple pumping illumination to the fiber **12B**. For this reason, the fiber **12B** will be referred to as the signal fiber while the fiber **12A** will be referred to as the pumping fiber.

The arcuate grooves **13A** and **13B** have a radius of curvature which is very large compared to the diameter of the fibers **12**, and have a width slightly larger than the fiber diameter to permit the fibers **12**, when mounted therein, to conform to a path defined by the bottom walls of the grooves **13**. The depth of the grooves **13A** and **13B** varies from a minimum at the center of the blocks **16A** and **16B**, respectively, to a maximum at the edges of the blocks **16A** and **16B**, respectively. This advantageously permits the fiber optic strands **12A** and **12B**, when mounted in the grooves **13A** and **13B**, respectively, to gradually converge toward the center and diverge toward the edges of the blocks **16A**, **16B**, thereby eliminating any sharp bends or abrupt changes in direction of the fibers **12** which may cause power loss through mode perturbation. In the embodiment shown, the grooves **13** are illustrated as being rectangular in cross section, however, it will be understood that other suitable cross sectional contours which will accommodate the fiber **12** may be used alternatively, such as a U-shaped cross section or a V-shaped cross section.

At the centers of the blocks **16**, in the embodiment shown, the depth of the grooves **13**, which mount the strands **12**, is less than the diameter of the strands **12**, while at the edges of the blocks **16**, the depth of the grooves **13** is preferably at least as great as the diameter of the strands **12**. Fiber optic material was removed from each of the strands **12A** and **12B** e.g. by polishing, to form the respective oval-shaped planar surfaces **18A**, **18B**, which are coplanar with the confronting surfaces **14A**, **14B**, respectively. These surfaces **18A**, **18B**, will be referred to herein as the fiber "facing surfaces". Thus, the amount of fiber optic material removed increases gradually from zero towards the edges of the block **16** to a maximum towards the center of the block **16**. This tapered removal of the fiber optic material enables the fiber cores to converge and diverge gradually, which is advantageous for avoiding backward reflection and excess loss of light energy.

In the embodiment shown, the coupler halves **10A** and **10B** are identical, and are assembled by placing the confronting surfaces **14A** and **14B** of the blocks **16A** and **16B** together, so that the facing surfaces **18A** and **18B** of the strands **12A** and **12B** are in facing relationship.

An index matching substance (not shown), such as index matching oil, is provided between the confronting surfaces **14**. This substance has a refractive index approximately equal to the refractive index of the cladding, and also functions to prevent the optically flat surfaces **14** from becoming permanently locked together. The oil is introduced between the blocks **16** by capillary action.

An interaction region **32** is formed at the junction of the strands **12**, in which light is transferred between the strands by evanescent field coupling. It has been found that, to insure proper evanescent field coupling, the amount of material removed from the fibers **12** must be carefully controlled so that the spacing between the core portions of the strands **12** is within a predetermined "critical zone". The evanescent fields extend into the cladding and decrease rapidly with distance outside their respective cores. Thus, sufficient material should be removed to permit each core to be positioned substantially within the evanescent field of the other. If too little material is removed, the cores will not be sufficiently close to permit the evanescent fields to cause the desired interaction of the guided modes, and thus, insufficient coupling will result. Conversely, if too much material is removed, the propagation characteristics of the fibers will be altered, resulting in loss of light energy due to mode perturbation. However, when the spacing between the cores of the strands **12** is within the critical zone, each strand receives a significant portion of the evanescent field energy from the other strand, and good coupling is achieved without significant energy loss. The critical zone is illustrated schematically in FIG. **5** are including that area, designated by the reference numeral **33**, in which the evanescent fields, designated by reference numerals **34A** and **34B**, of the fibers **12A** and **12B**, respectively, overlap with sufficient strength to provide coupling, i.e., each core is within the evanescent field of the other.

The blocks or bases **12** may be fabricated of any suitable rigid material. In one presently preferred embodiment, the bases **12** comprise generally rectangular blocks of fused quartz glass approximately one inch long, one inch wide, and 0.4 inch thick. In this embodiment, the fiber optic strands **12** are secured in the slots **13** by suitable cement **38**, such as epoxy glue. One advantage of the fused quartz blocks **16** is that they have coefficient of thermal expansion similar to that of glass fibers, and this advantage is particularly important if the blocks **16** and fibers **12** are subjected to any heat treatment during the manufacturing process. Another advantage of the fused quartz blocks is that, as they are made of the same material as the optical fiber, they polish at the same rate as the optical fiber, and thus, provide a continuous support to the fiber during polishing. Other suitable materials for the block **16** include silicon, which also has excellent thermal properties for this application.

OPERATION OF THE COUPLER 10

The coupler **12** includes four ports, labeled A, B, C, and D in FIG. **1**. When viewed from the perspective of FIG. **1**, ports A and C, which correspond to strands **12A** and **12B**, respectively, are on the left-hand side of the coupler **10**, while the ports B and D, which correspond to the strands **12A** and **12B**, respectively, are on the right-hand side of the coupler **10**. For the purposes of discussion, it will be assumed that input light is applied to port A. This light passes through the coupler and is output at port B and/or port D, depending upon the amount of power that is coupled between the strands **12**. In this regard, the term "normalized coupled power" is defined as the ratio of the coupled power to the total output power. In the above example, the normalized coupled power would be equal to the ratio of the power at port D to the sum of the power output at ports B and D. This ratio is also referred to as the "coupling efficiency", and when so used is typically expressed as a percent. Thus, when the term "normalized coupled power" is used herein, it should be understood that the corresponding coupling efficiency is equal to the normalized coupled power times **100**. In this regard, tests have shown that the coupler **10** has a coupling

efficiency of up to 100%. However, it will also be seen that the coupler **10** may be "tuned" to adjust the coupling efficiency to any desired value between zero and the maximum. Such tuning may be accomplished e.g. by relatively laterally sliding the fibers in a direction perpendicular to their length. The relative positions of the fibers may be defined in terms of their offset, i.e. the distance between the central axes of the fiber cores, measured in the direction perpendicular to their length, along the surfaces **14**. Thus, when the oval surfaces **18** are superimposed, the offset is zero, and the offset increases as the fibers **12** are laterally separated by relatively sliding the blocks **16**.

The coupler **10** is highly directional, with substantially all of the power applied at one side of the coupler being delivered to the other side of the coupler. The coupler directivity is defined as the ratio of the power at port D to the power at port C, with the input applied to port A. Tests have shown that the directionally coupled power (at power D) is greater than **60** db above the contra-directionally coupled power (at port C). Further, the coupler directivity is symmetrical. That is, the coupler operates with the same characteristics regardless of which side of the coupler is the input side and which side is the output side. Moreover, the coupler **10** achieves these results with very low throughput losses. The throughput loss is defined as the ratio of the total output power (ports B and D) to the input power (port A) subtracted from one (i.e., $1-(P_B+P_D)/P_A$). Experimental results show that throughput losses of 0.2 db have been obtained, although losses of 0.5 db are more common. Moreover, these tests indicate that the coupler **10** operates substantially independently of the polarization of the input light applied.

The coupler **10** operates on evanescent field coupling principles in which guided modes of the strands **12** interact, through their evanescent fields, to cause light to be transferred between the strands **12**. As previously indicated, this transfer of light occurs at the interaction region **32**. The amount of light transferred is dependent upon the proximity and orientation of the cores, as well as the effective length of the interaction region **32**. As will be described in detail below, the amount of light transferred is also dependent of the wavelength of the light. The length of the interaction region **32** is, in turn, dependent upon the radius of curvature of the fibers **12**, and, to a limited extent, the core spacing, although it has been found that the effective length of the interaction region **32** is substantially independent of core spacing. However, the "coupling length", i.e., the length within the interaction region **32** which is required for a single, complete transfer of a light signal from one fiber **12** to the other, is a function of core spacing, as well as wavelength. In one exemplary embodiment, employing an edge-to-edge core spacing of about **1.4** microns, and a radius of curvature on the order of 25 centimeters, the effective interaction region is approximately one millimeter long at a light signal wavelength of 633 nm. Because the coupling length at 633 nm is also one millimeter in such a coupler, the light makes only one transfer between the strands **12** as it travels through the interaction region **32**. However, if the length of the interaction region **32** is increased, or core spacing decreased, a phenomenon referred to herein as "overcoupling" will occur, since the coupling length is shorter than the effective interaction length. Under these circumstances, the light will transfer back to the strand from which is originated. As the interaction length is further increased, and/or the core spacing further decreased, the effective interaction length increases relative to the coupling length, and at least some of the light transfers back to the other strand. Thus, the light may make multiple transfers back and forth between the two strands **12** as it travels through the region **32**, the number of such transfers being dependent on the length of the interaction region **32**, the light wavelength (as described below), and the core spacing.

Since the coupling length in a single mode fiber coupler, as described above, shows a strong dependence on signal wavelength, as described in detail in the copending application incorporated above, it is possible with properly chosen geometrical parameters for the coupler **10**, to totally couple one signal wavelength while a second signal wavelength remains essentially uncoupled. This phenomenon permits the combination of two signals fed into the ports on one side of the coupler **10**. Thus, as shown in FIG. **1**, if a pumping signal having a wavelength λ_1 is fed into port A of coupler **10**, and a signal to be amplified, having a wavelength λ_2 is coupled to port C, and the geometry is properly selected, both signals can be combined at port D, with virtually no light output at port B.

To illustrate this wavelength dependence, FIG. **6** provides a plot of coupled power versus signal wavelength in the visible and near infrared spectrum for a particular coupler geometry. Because for this coupler configuration the effective interaction length of the coupler is an odd multiple of the coupling length for the wavelength 720 nm, but an even multiple of the coupling length for the wavelength 550 nm, the wavelength 720 nm will be 100% coupled, while the wavelength 550 nm will be effectively uncoupled. With different efficiencies, different wavelengths may be combined or separated. For instance, 590 nm and 650 nm may be separated or combined at an 80% efficiency.

Virtually any pair of wavelengths (λ_1, λ_2) may be efficiently combined or separated so long as the effective interaction length is an even multiple of the coupling length for one wavelength and an odd multiple for coupling length for the other wavelength. As the number of coupling lengths within the effective interaction length increases, the resolution of the multiplexer is enhanced. As is described in detail in the incorporated reference, the multiplexer resolution may be enhanced by increasing the radius of curvature of the fibers **12A,12B**. Provided that the interaction length of the coupler is large enough, virtually any two signals may be exactly mixed or separated, regardless of how closely spaced their wavelengths are.

The interaction length is a function of wavelength, and the resolution is approximately proportional to $(R)^{\frac{1}{2}}$. As R increases, the effective interaction length increases, and becomes a higher multiple of the coupling length, improving resolution. This result is illustrated in FIG. **7**, which is comparable to the graph of FIG. **6**, except that the radius of curvature has been increased to 200 centimeters. As anticipated, this increase in radius improves the coupler resolution near $g=600$ nm from approximately 170 nm in the 25 centimeter radius example of FIG. **6** to approximately 60 nm in the 200 centimeter case.

The resolution of a multiplexing coupler depends on two independent parameters, H (fiber spacing) and R (radius of curvature of the fibers). For a given pair of signal wavelengths, efficient mixing may be achieved by first properly selecting a fiber spacing H for the

coupler which yields a large wavelength dependence for the wavelengths of interest (choice of H), and then by selecting a radius of curvature which yields a resolution equal to the difference between the wavelengths (choice of R).

After the resolution of the coupler has been set in accordance with the wavelengths to be separated, the coupler may be tuned to precisely adjust the coupling lengths for the wavelengths of interest so that the effective interaction length is an even multiple of the coupling length of one wavelength and an odd multiple of the coupling length of the other wavelength. This is accomplished by offsetting the fibers by sliding the blocks **16A,16B** (FIG. **1**) relative one another in a direction normal to the axis of the fibers **12A,12B**. Such an offset has the effect of increasing the minimum fiber spacing H and increasing the effective radius of curvature of the fibers. If the required offset is small enough, it will not upset the multiplexer resolution. This stems from the fact that the separation H of a large radius coupler changes rapidly with fiber offset in comparison to changes in the effective radius of curvature with fiber offset.

To illustrate this tunability of multiplexing couplers, FIG. **8** provides a plot of relative coupled power versus wavelength for three increasing values of fiber offset (0 microns, 0.5 microns, and 1.0 microns). The curve is seen to shift toward increasing wavelengths as the offset increases, while the period of oscillation (or resolution) remains virtually unchanged. In this particular example (R=200 cm, H=4 microns), a 1-micron offset shifts the curve by approximately 45 nm.

OVERALL OPERATION OF THE AMPLIFIER

Referring now to FIG. **9**, the manner in which the amplifier of the present invention utilizes the wavelength multiplexing properties of the coupler **10** to provide pumping illumination to energize ND:YAG fiber will be described.

A source of pumping illumination **42** is coupled to the fiber **12A** of the coupler **10** to provide pumping illumination at port A of the multiplexing coupler **10**, and a signal to be amplified is coupled to one end of the fiber **12B** at port C of the coupler **10**. The pumping illumination from the source **42** and the signal to be amplified are combined at port D at the coupler **10** through the multiplexing action of the coupler, as explained above. Thus, the coupler **10** is adjusted to have a 100% coupling efficiency at the wavelength of the source **42**, and a 0% coupling efficiency, at the wavelength of the signal input at port C. This pair of signal wavelengths is supplied to an ND:YAG crystal **44** which is coupled to the fiber **12B** in the manner described below. The signal input at the port C will be amplified in the ND:YAG crystal **44** and the amplified signal will be coupled from this crystal **44** to an optical fiber **46** for transmission within the optical fiber system. The diameter of the ND:YAG crystal **44** may be extremely small in comparison with the diameter of ND:YAG rods used in prior art amplifiers. For example, an amplifier has been constructed in which the crystal **44** has a diameter of **100** microns. Even smaller diameters are feasible, approaching the diameter of the single mode fiber **12B**. Coupling between the fiber **12B** and the crystal **44** is enhanced as the diameter of the crystal **44** is reduced and signal gain is increased since the density of pumping illumination from the source **42** within the crystal **44** increases as the crystal **44** diameter is reduced.

ND:YAG AMPLIFICATION

Referring now to FIG. **10**, which is a diagram of the absorption spectrum of the ND:YAG crystal **44** at 300° K., it can be seen that ND:YAG material has a relatively high optical density, and thus a short absorption length, at selected wavelengths. For this reason, it is advisable to select the wavelength of the pumping illumination source **42** in order to permit the absorption length to be as short as possible. This will permit substantially complete absorption of the pumping illumination within a very short length of the ND:YAG crystal **44**. As can be seen from FIG. **10**, the wavelength 0.58 microns is best suited for pumping illumination, although the wavelengths 0.75 and 0.81 microns are relatively well suited.

Referring now to FIG. **11A**, which is an energy level diagram for the ND:YAG crystal **44**, it will be understood that, when pump light at the absorption wavelength, described above, is absorbed by the ND:YAG crystal **44**, the neodymium ions are excited from the ground state to the pump band. From the pump band, the ions quickly relax, through phonon interactions, to the upper lasing level. From this upper lasing level, the neodymium ions will undergo a relatively slow fluorescence to the lower lasing level. From this latter level, a final, rapid phonon relaxation occurs to the ground state. This latter rapid relaxation in a four-level laser system of the type shown in FIG. **11A** is advantageous, since the rapid phonon relaxation between the lower lasing level and the ground state provides a practically empty lower lasing level. This feature is shown in FIG. **11B**, in which the population densities at the pump band upper lasing level, lower lasing level, and ground state are shown for the ND:YAG crystal **44** during continuous pumping. Because the rage of fluorescence between the upper lasing level and lower lasing level is relatively slow in comparison with the phonon relaxation between the pump band and the upper lasing level, as well as between the lower lasing level and the ground state, the population density at the upper lasing level is substantially higher than that at the lower lasing level, yielding a high inversion ratio. The average lifetime of neodymium ions at the upper lasing level, prior to spontaneous fluorescence, is about 230 microseconds at 300° k. in ND:YAG.

The signal to be amplified is selected to have a wavelength at the laser transition wavelength (1.064 microns), i.e., the wavelength of light emitted by the ND:YAG ions during relaxation between the upper and lower lasing levels. When this signal is supplied to the crystal **44** by the coupler **10** (FIG. **9**), it will trigger the emission of stimulated photons at the same frequency, coherent with the signal, and the signal is thereby amplified. Thus, the passage of light at this frequency will cause a photon emitting relaxation between the upper lasing level and lower lasing level of FIG. **11A**, in phase with the light signal to be amplified, yielding an effective gain for the input light signal.

The gain which can be achieved in the amplifier of this invention is dependent upon the density of the inverted neodymium ion population within the ND:YAG crystal **44**. Initially, the ultimate inversion population is limited by the lattice structure of the YAG material **44** itself since the ND:YAG material, some yttrium atoms are replaced with neodymium atoms in the crystal lattice. Only approximately 1 yttrium atom in each 100 yttrium atoms may be replaced by a neodymium ion

without distorting the lattice structure on the ND:YAG material.

Theoretical calculations of the small gain signal (g_0 of the amplifier of this invention can be made, using the relation $g_0\sigma\Delta N$, where σ is the stimulated emission cross section, for ND:YAG, $8.8\times10^{-19}cm^2$, and ΔN is the population inversion density given by:

$$\Delta N = \frac{P_p}{V}\ \frac{\eta_1\eta_2 t_{sp}}{h\nu} \quad (1)$$

where P_p is the absorbed pump power, V is the crystal volume and thus, P_p/V is the absorbed pump power per unit of fiber volume, t_{sp} is the spontaneous radiative lifetime, that is, the 230-microsecond fluorescence relaxation time of the neodymium ions, η_1 is the effective spectral overlap of pump output with an ND:YAG absorption line, as shown in FIG. **10**, η_2 is equal to the quantum efficiency of 1.06-micron fluorescence, namely 0.63, and hy is equal to the energy of one pump photon.

Combining the above relationship provides:

$$g_0 = \sigma\frac{P_p}{V}\ \frac{\eta_1\eta_2 t_{sp}}{h\nu} \quad (2)$$

for the dependence of gain on pump power. It should be recognized that the value P_p is the absorbed pump power and that an increase in the length of the crystal **44** does not necessarily increase the gain. Thus, if the pumping radiation from the source **42** is coupled completely to the ND:YAG crystal **44**, and travels in the crystal **44** a distance which is sufficient to permit this crystal **44** to nearly completely absorb the pumping radiation, then the value P_p in this equation may be replaced by the input power level. To obtain the net gain, however, one must subtract from 9_0 the propagation losses within the ND:YAG crystal **44** at 1.06 microns. A loss of 100 db per kilometer would reduce the gain by only 0.001 dB per centimeter. Thus, if the overall length of the crystal **44** can be maintained relatively short, while still absorbing substantially all of the input pump power, the propagation losses within the amplifier can be maintained at a low level.

DETAILED OPERATION OF THE AMPLIFIER

Referring again to FIG. **30**, the pumping source **42** coupled to the fiber **12A** at the port A of the coupler **10**, through the multiplexing action of the coupler **10**, provides pumping illumination for the ND:YAG crystal **44**. The pumping source **42** may be, for example, a long-life LED, such as those currently available which operate at a current density of approximately 1,000 amps per cm^2 and have a radiance of approximately 5 watts per $sr.cm^2$. In fact, some LEDs have been reported with a radiance of approximately 50 watts/sr./cm^2. Because of the size differential between the single mode fiber **12A** and these LEDs, a lens may be useful in focusing the output of the LED source into the fiber **12A**.

Alternatively, the pump source **42** may be laser diode which permits even higher concentrations of pump power in the fiber **12A** and thus in the ND:YAG crystal **44**.

Regardless of the type of pumping source **42** utilized, the efficiency of the system will be enhanced if the wavelength of the radiation from this source **42** corresponds to a peak in the absorption spectrum of the ND:YAG crystal **44**, shown in FIG. **10**. Electroluminescent diodes are commercially available with appropriate dopings to emit spectra in the 0.8 micron range which match quite well the absorption spectrum of room temperature ND:YAG material. For example, commercially available GaAlAs LEDs provide radiation spectra which are strong at the 0.8 micron region. Similarly, laser diode structures are commercially available which emit energy over the 0.8 to 0.85 micron range. In addition, the pump wavelength should be as close to the signal wavelength as allowed by the spectroscopy of the ND:YAG, to maximize the overall pumping efficiency.

It will be recalled that the lasing frequency of the ND:YAG material of the crystal **44** is 1.06 micron. The multiplexing coupler **10** is thus fabricated for use in this invention to provide virtually complete coupling at the wavelength of the pumping source **42**, 0.8 microns in the above example, while providing substantially no coupling at the lasing frequency of the ND:YAG crystal **44**, 1.06 microns in this same example.

This selective coupling is accomplished, in accordance with the techniques described above, by properly selecting the fiber spacing H to yield a large wavelength dependence for wavelengths between 0.8 microns and 1.06 microns, and then by selecting a radius of curvature for the fibers **12A,12B** which yields a resolution equal to the difference between 1.06 and 0.8 microns, or 0.26 microns. After the resolution of the coupler has been set in this manner, the coupler may be tuned, as previously described, to adjust the coupling length for the wavelengths 0.8 microns and 1.06 microns so that the effective interaction length is an even multiple of the coupling length for one of this pair of wavelengths and an odd multiple of the coupling length for the remaining wavelength. In the example shown in FIG. **9**, since it is desired to couple the output of the pump source **42** into the fiber **12B**, the effective interaction length for the coupler should be adjusted to be an odd multiple of the coupling length at the wavelength of the pump source **42**, i.e., 0.8 microns, and to be an even multiple of the signal frequency 1.06 microns. This will result in a complete coupling of the illumination from the pump source **42**, from the fiber **12A** into the fiber **12B**, with essentially no coupling of the signal to be amplified from the fiber **12B** to the fiber **12A**. It will be understood, of course, that no coupling in this instance means an even number of complete couplings so that, for example, if the effective interaction length at the region **32** is twice the coupling length at 1.06 microns, the signal to be amplified will be coupled two complete times, once from the fiber **12B** to the fiber **12A**, and then from the fiber **12A** to the fiber **12B**. If this signal enters the coupler at port C, as shown on the left of FIG. **9**, it will exit uncoupled at port D. However, at port D, the signal to be amplified will coexist with light from the pumping source **42**, which will be completely coupled from the fiber **12A** to the fiber **12B**.

Since light from the pumping source **42** will be transmitted along the fiber **12B**, after coupling, this pumping illumination will invert the neodymium ions in the ND:YAG crystal **44**. Thus, a signal, which is injected at port C and exits, uncoupled, from port D, will be amplified in the manner previously described as it passes through the crystal **44**, since this signal will excite spontaneous lasing relaxation of the ND:YAG material of the crystal **44**, which lasing relaxation will provide light coherent with the signal to be amplified.

The amplifier of the present invention therefore provides a convenient means to transfer pumping illumination from the pump source **42** by wavelength dependent coupling to the ND:YAG crystal **44**, while prohibiting coupling of the signal to be amplified from the fiber **12B** to the fiber **12A**. It should be recognized that the results achieved in the present invention may also be realized using a coupler in which the coupling efficiency at the pumping illumination wavelength is 0%, while the coupling efficiency at the light signal wavelength is 100%. In this case, the pumping source would be coupled to port C of the coupler **10**, while the input light signal to be amplified would be coupled to port A.

BIDIRECTIONAL SYMMETRY

In order to make the amplifier symmetrically bidirectional, a pair of pump sources **42,48** should be utilized, as shown in FIG. **12**, along with a pair of multiplexing couplers **50,52**. It will be understood that, if such bidirectional symmetry is not necessary, either of the pump sources **42,48** will invert ions at one end of the ND:YAG crystal **44** and will thus yield gain for signals transmitted in either direction in the crystal **44**.

If only one of the pump sources **42,48** is utilized, it should be recognized that the ND:YAG crystal **44** will not be uniformly illuminated. Thus, the inverted population of neodymium ions will not be uniformly distributed along the length of the crystal **44**. Because this non-uniform or non-symmetrical state within the amplifier may yield different gain for signals input at the fiber **54**, than for signals input at the fiber **56** (particularly when these signals occur simultaneously), it is advantageous to utilize the pair of sources **42,48**.

The phenomenon of dissimilar gain for signals traversing the fiber crystal **44** in different directions with a non-symmetrical inversion population of neodymium ions occurs as follows. It will recognized that, as a signal to be amplified propagates from left to right in the crystal **44** of FIG. **12**, it will trigger the emission of stimulated photons within the ND:YAG crystal **44**. Such triggering emission, of course, lowers the inversion population within the crystal **44**. If, for example, in a gyroscope, a pair of waves propagate simultaneously through the crystal **44** in opposite directions, the signal input at the left end will deplete the inversion population adjacent this end before the signal input at the right end arrives at the left end of the crystal **44**, as viewed in FIG. **12**. If the inversion population is higher at the left end of the crystal **44**, than at the right end, as would be the case if only the pump source **42** were utilized, the signal input at the left will undergo a greater amplification, since it will deplete the inversion population before the signal which is input at the right end arrives at the high density left end.

It should be recognized that the pumping illumination supplied by the pump sources **42,48** should be sufficient, on a continuing basis, to replace the depleted population within the crystal **44** which occurs when the signals are amplified. Thus, for example, in a gyroscope where a pulse signal circulates through a kilometer of fiber, a counter-propagating signal will traverse the amplifier, shown in FIG. **12**, approximately once each **5** microseconds. If continuous pump sources **42,48** are used, they should provide sufficient output so that, during each **5**-microsecond period, they are capable of reinverting the neodymium ion population which is relaxed during each successive traverse of the signals to reinvert a population equal to that which has relaxed, such that the amplification factor or gain of the amplifier will remain relatively constant.

As will be recognized from the above-description, a proper selection of fiber spacing and radius of curvature will yield a coupler which permits pumping sources **42,48** to illuminate the crystal **44** to invert the neodymium population therein. With a proper selection of the coupler parameters, the signal to be amplified is not coupled from the fibers **54,56**, and thus traverses the crystal **44** to be amplified by stimulating relaxation of neodymium ions in the crystal **44** which produces light coherent with the signal to be amplified.

SIGNAL INSERTION

Referring now to FIG. **13**, a system arrangement is shown in which the multiplexing properties of a pair of couplers **58,60** is used to permit a pumping source **42** to illuminate the ND:YAG crystal **44** while also permitting an injection of signals into a recirculating fiber optic system. In this instance, for example, it may be assumed that the fiber ends **62,64** are a part of a recirculating fiber loop and that the system, in addition to supplying pumping illumination for the crystal **44**, to amplify signals circulating in this loop, must supply signals for injection into the loop.

In the system shown in FIG. **13**, the coupler **58** is arranged to couple light wavelengths from the illumination source **42** with a coupling efficiency of 100%, while leaving the input signal at an input fiber **66** effectively uncoupled. This will yield a combination signal on the fiber segment **68** which includes both the pumping illumination from the source **42** and the input signal from the fiber **66**. If the characteristics of the coupler **60** are properly selected in the manner described above, and this coupler **60** is properly tuned, the coupler **60** may be arranged to couple 100% of light wavelengths from the pumping signal **42** but only 1% of wavelengths at the signal input from the fiber segment **68**. Thus, 1% of the input signal from the fiber segment **66** will be injected into the fiber recirculating loop and will initially be transmitted through the crystal **44** to the fiber segment **64** for recirculation. At the same time, 100% of the pumping illumination from the source **42** will be injected into the end of the crystal **44** to provide amplification for the recirculating signal.

When this recirculating signal appears at the fiber segment **62**, only 1% of the signal will be lost at the coupler **60** and this 1% may supply an output for a sensor **70** to monitor the recirculating light. The remaining **99%** of the recirculating light signal will be injected into the crystal **44** from the coupler **60**, for reamplification and recirculation within the fiber loop. This system, therefore, with the use of two couplers **58,60**, provides continuous sensing of the light circulating in the fiber loop, signal injection into the loop, and continuous pumping for the amplifying crystal **44**. The coupling efficiencies provided above are exemplary only, but it should be recognized that the coupling efficiency of the coupler **60** should be maintained relatively low at the wavelength of the signal propagating in the loop so that only a small portion, smaller than the amplification added to the signal at the crystal **44**, is subtracted from the recirculating signal by the coupler **60**.

SUMMARY

The combination of the multiplexing coupler with an ND:YAG amplifying crystal permits end pumping of

the amplifying crystal with simultaneous signal injection at the end of this fiber so that the pumping illumination need not be carefully timed with respect to the signal to be amplified. Symmetrical bidirectional amplification is possible and signal injection using the same multiplexing coupler may be achieved.

What is claimed is:

1. A fiber optic amplifier, comprising:

an optical fiber comprised of a material which will emit light in response to pumping;

a source of signals comprising light at a frequency of said emitted light;

a source of light for pumping said material; and

means for coupling both said signals and said light for pumping into an end of said optical fiber.

2. A fiber optic amplifier as defined in claim **1** in which said means for coupling comprises an optical coupler.

3. A fiber optic amplifier as defined in claim **2**, wherein said optical coupler comprises a single-mode optical coupler.

4. A fiber optic amplifier as defined in claim **2** in which said optical coupler utilizes evanescent field coupling.

5. A fiber optic amplifier as defined in claim **3** in which said optical coupler provides different coupling coefficients for said signals and said light for pumping.

6. A fiber optic amplifier as defined in claim **5** in which said optical coupler has a coupling efficiency which is wavelength dependent and in which said signals and said light for pumping are at different wavelengths, yielding different coupling efficiencies for said signals and said light for pumping.

7. A fiber optic amplifier, comprising:

a fiber optic coupler including a pair of optical fibers juxtaposed to provide coupling of light at a first frequency between said fibers and to prohibit coupling of light at a second frequency between said fibers;

a source of pumping illumination coupled to a first end of one of said pair of fibers, said pumping illumination comprising light having said first frequency;

a source of a signal to be amplified, coupled to a first end of the other of said pair of fibers, said signal to be amplified comprising light having said second frequency; and

a laser fiber formed of material which will possess a laser transition at said second frequency of said signal to be amplified if said material is pumped with said pumping illumination, said laser fiber coupled at one end to a second end of said other of said pair of fibers.

8. A fiber optic amplifier as defined in claim **7** in which said fiber optical coupler has an effective interaction length at the juxtaposition of said optical fibers which is an even multiple of the coupling length of said fibers at said juxtaposition at a first wavelength corresponding to one of said first and second frequencies and an odd multiple of the coupling length of said fibers at said juxtaposition at a second wavelength corresponding to the other of said first and second frequencies.

9. A fiber optical amplifier as defined in claim **8**, wherein said pair of optical fibers are laterally offset from one another to tune said coupler to provide said odd and even multiples.

10. A fiber optic amplifier as defined in claim **9**, wherein said pair of optical fibers are arcuate and wherein the radius of said arcuate optical fibers is selected in accordance with the difference between said first and second wavelengths.

11. A fiber optic apparatus, comprising:

a source of pump light comprising light having a first wavelength;

an optical fiber comprising a laser material, said optical fiber emitting light at a second wavelength in response to pumping at said first wavelength; and

an optical coupler having an input port and an output port, said input port coupled to said pump source to receive light from said pump source, said output port coupled to said optical fiber for pumping said laser material, said coupler being highly wavelength discriminating so as to selectively couple one of said first and second wavelengths without substantial coupling of the other of said first and second wavelengths.

12. A method for amplifying a light signal carried by an optical fiber, comprising the steps of:

combining said light signal and pumping illumination on a single optical fiber; and

coupling said combined light signal and pumping illumination from said single optical fiber to one end of a fiber comprised of a material which will emit stimulated radiation at a wavelength of said light signal if pumped with said pumping illumination.

13. A method of amplifying a light signal as defined in claim **12**, wherein said combining step comprises multiplexing said wavelength of said light signal and a wavelength of said pumping illumination in an optical coupler which is optically connected to said single optical fiber and which has a coupling efficiency which is wavelength dependent.

14. A method of amplifying a light signal as defined in claim **13**, wherein said multiplexing step comprises:

juxtaposing a pair of optical fibers to provide an interaction length; and

applying said light signal to one of said fibers and said pumping illumination to the other of said fibers.

15. A method of amplifying a light signal as defined in claim **12**, wherein the step of coupling comprises coupling said combined light signal and pumping illumination to a laser fiber having a diameter which is less than the absorption length of said laser fiber at said wavelength of said pumping illumination.

* * * * *

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REEXAMINATION CERTIFICATE (4193rd)

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[11] **B1 4,859,016**

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[54] **FIBER OPTIC AMPLIFIER**

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[*] Notice: This patent is subject to a terminal disclaimer.

Related U.S. Application Data

[62] Division of application No. 06/930,136, Nov. 12, 1986, Pat. No. 4,723,824, which is a division of application No. 06/554,888, Nov. 25, 1983, Pat. No. 4,674,830.

[51] **Int. Cl.**[7] **G02B 6/26**; H01S 3/067

[52] **U.S. Cl.** **385/5**; 359/341; 359/342; 372/6; 372/69

[58] **Field of Search** 385/4, 5, 15, 27, 385/28, 30, 31, 38, 39, 122, 126, 127; 372/3, 6, 69, 70, 71, 94; 359/333, 341, 342, 343, 345

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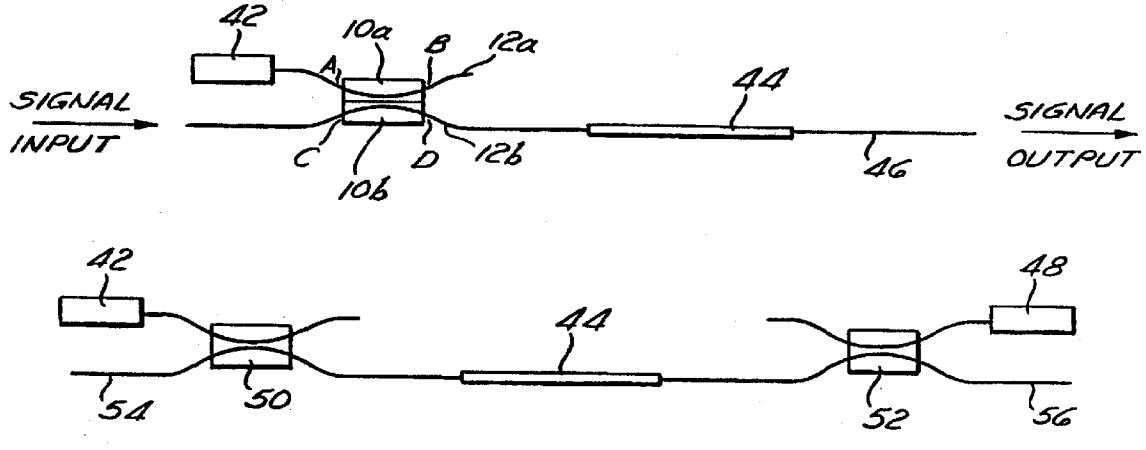
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(List continued on next page.)

Primary Examiner—John D. Lee

[57] **ABSTRACT**

An amplifier for use with fiber optic systems comprises a neodymium YAG crystal placed in series with a signal-carrying optical fiber. The ND:YAG crystal is supplied by the optical fiber with both the signal to be amplified, and pumping illumination. The pumping illumination is coupled onto the optical fiber by a multiplexing coupler which is used to combine the signal to be amplified and illumination from a pumping illumination source onto a single optical fiber. The pumping illumination inverts the neodymium ions within the ND:YAG crystal. The signal to be amplified propagates through this crystal to stimulate emission of coherent light from the neodymium ions, resulting in amplification of the signal. Because this arrangement permits the ND:YAG crystal to be end-pumped with pumping illuminations, and because the length of the ND:YAG crystal may be substantially greater that the absorption length for the crystal at the wavelength of the pumping illumination, virtually all of the pumping illumination may be absorbed within the ND:YAG crystal and used for amplification of the signal carried by the optical fiber.



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REEXAMINATION CERTIFICATE ISSUED UNDER 35 U.S.C. 307

THE PATENT IS HEREBY AMENDED AS INDICATED BELOW.

Matter enclosed in heavy brackets [ ] appeared in the patent, but has been deleted and is no longer a part of the patent; matter printed in italics indicates additions made to the patent.

AS A RESULT OF REEXAMINATION, IT HAS BEEN DETERMINED THAT:

The patentability of claims **7–15** is confirmed.

Claims **1**, **2**, **5** and **6** are cancelled.

Claims **3** and **4** are determined to be patentable as amended.

New claims **16–22** are added and determined to be patentable.

3. A fiber optic amplifier [as defined in claim **2**, wherein] *comprising:*

an optical fiber comprised of a material which will emit light in response to pumping;

a source of signals to be amplified comprising light at a frequency of said emitted light;

a source of light for pumping said material; and

means for coupling both said signals and said light for pumping into an end of said optical fiber, said means for coupling comprising an optical coupler, said optical coupler [comprises] *being highly wavelength discriminating and comprising* a single-mode optical coupler *having a plurality of input/output ports comprised of respective single mode fiber portions, one of said ports being coupled to said optical fiber so as to transmit both pump light and signals to be amplified between said optical coupler and said material which will emit light in response to pumping, a second of said ports being coupled to transmit said pump light from the pump source to said optical fiber, and a third of said ports transmitting said signals to be amplified.*

4. A fiber optic amplifier [as defined in claim **2** in which said], *comprising:*

an optical fiber comprised of a material which will emit light in response to pumping;

a source of signals comprising light at a frequency of said emitted light;

a source of light for pumping said material; and

means for coupling both said signals and said light for pumping into an end of said optical fiber, said coupling means comprising an optical coupler *which* utilizes evanescent field coupling.

16. The fiber optic apparatus of claim 11, wherein said source of light for pumping is coupled to said optical fiber such that said pump light propagates through said laser material in a propagation direction to pump said laser material, whereby a population inversion is created in the laser material, and wherein said optical fiber receives signals to be amplified for propagation in the same propagation direction as said light for pumping, whereby relaxation of the population inversion in said laser material amplifies said signals as said signals propagate in said propagation direction.

17. The fiber optic apparatus of claim 11, wherein said optical fiber has a first propagation direction and a second propagation direction opposite said first propagation direction, and wherein said source of pump light is coupled to said optical fiber such that said pump light propagates through said laser material in at least said first propagation direction to pump said laser material, whereby a population inversion is created in said laser material, and wherein said optical fiber receives signals to be amplified for propagation in at least said second direction, whereby relaxation of the population inversion in said laser material amplifies said signals as said signals propagate in said second propagation direction.

18. The fiber optic apparatus of claim 17, wherein said apparatus comprises a second source of pump light coupled to said optical fiber such that pump light from the second source propagates through said laser material in said second direction.

19. The fiber optic apparatus of claim 11, wherein said optical fiber has a diameter that at least approaches single mode to increase pump light density and thereby increase population inversions in said laser material as said pump light propagates within said diameter through said laser material.

20. The fiber optic apparatus of claim 11, wherein the optical fiber is formed substantially entirely of said laser material.

21. The fiber optic apparatus of claim 11, wherein said ports comprise respective portions of single mode optical fiber.

22. The fiber optic apparatus of claim 21, wherein said optical coupler utilizes evanescent field coupling.

* * * * *

US004859016C2

(12) EX PARTE REEXAMINATION CERTIFICATE (6821st)
United States Patent
Shaw et al.

(10) Number: US 4,859,016 C2
(45) Certificate Issued: May 12, 2009

(54) **FIBER OPTIC AMPLIFIER**

(75) Inventors: **Herbert J. Shaw**, Stanford, CA (US); **Michel J. F. Digonnet**, Palo Alto, CA (US)

(73) Assignee: **The Board of Trustees of the Leland Stanford Junior University**

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No. 90/006,726, Jul. 24, 2003
No. 90/007,147, Jul. 30, 2004

Reexamination Certificate for:
Patent No.: **4,859,016**
Issued: **Aug. 22, 1989**
Appl. No.: **07/137,860**
Filed: **Dec. 22, 1987**

Reexamination Certificate B1 4,859,016 issued Nov. 7, 2000

Related U.S. Application Data

(62) Division of application No. 06/930,136, filed on Nov. 12, 1986, now Pat. No. 4,723,824, which is a division of application No. 06/554,888, filed on Nov. 25, 1983, now Pat. No. 4,674,830.

(51) **Int. Cl.**
G02B 6/26 (2006.01)
H01S 3/067 (2006.01)

(52) **U.S. Cl.** **385/5**; 359/341.3; 359/342; 359/349; 372/6; 372/69

(58) **Field of Classification Search** 385/4, 385/5, 15, 27, 28, 30, 31, 38, 39, 122, 126, 385/127, 42; 372/3, 6, 69, 70, 71, 94; 359/333, 359/341, 342, 343, 345
See application file for complete search history.

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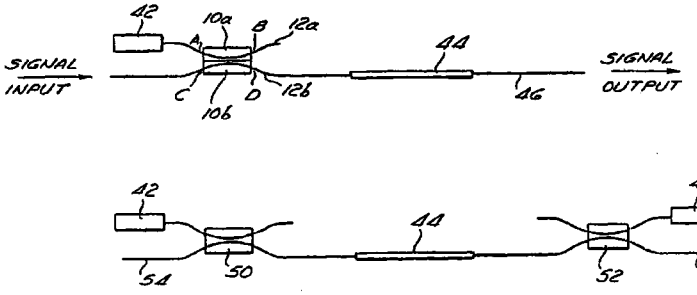
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Primary Examiner—Margaret Rubin

(57) **ABSTRACT**

An amplifier for use with fiber optic systems comprises a neodymium YAG crystal placed in series with a signal-carrying optical fiber. The ND:YAG crystal is supplied by the optical fiber with both the signal to be amplified, and pumping illumination. The pumping illumination is coupled onto the optical fiber by a multiplexing coupler which is used to combine the signal to be amplified and illumination from a pumping illumination source onto a single optical fiber. The pumping illumination inverts the neodymium ions within the ND:YAG crystal. The signal to be amplified propagates through this crystal to stimulate emission of coherent light from the neodymium ions, resulting in amplification of the signal. Because this arrangement permits the ND:YAG crystal to be end-pumped with pumping illuminations, and because the length of the ND:YAG crystal may be substantially greater than the absorption length for the crystal at the wavelength of the pumping illumination, virtually all of the pumping illumination may be absorbed within the ND:YAG crystal and used for amplification of the signal carried by the optical fiber.

At the time of issuance and publication of this certificate, the patent remains subject to pending reexamination control numbers 90/008,744 and 90/008,941 filed Sep. 20, 2007 and Nov. 28, 2007 respectively. The claim content of the patent may be subsequently revised in the reexamination proceedings.



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EX PARTE REEXAMINATION CERTIFICATE ISSUED UNDER 35 U.S.C. 307

NO AMENDMENTS HAVE BEEN MADE TO THE PATENT

AS A RESULT OF REEXAMINATION, IT HAS BEEN DETERMINED THAT:

The patentability of claims **3**, **4** and **7**–**22** is confirmed.

Claims **1**, **2**, **5** and **6** were previously cancelled.

* * * * *

US004859016C3

(12) EX PARTE REEXAMINATION CERTIFICATE (7230th)

United States Patent

Shaw et al.

(10) Number: US 4,859,016 C3

(45) Certificate Issued: Dec. 8, 2009

(54) **FIBER OPTIC AMPLIFIER**

(75) Inventors: **Herbert J. Shaw**, Stanford, CA (US); **Michel J. F. Digonnet**, Palo Alto, CA (US)

(73) Assignee: **The Board of Trustees of the Leland Stanford Junior University**

Reexamination Request:

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Filed: **Dec. 22, 1987**

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Reexamination Certificate C2 4,859,016 issued May 12, 2009

Related U.S. Application Data

(62) Division of application No. 06/930,136, filed on Nov. 12, 1986, now Pat. No. 4,723,824, which is a division of application No. 06/554,888, filed on Nov. 25, 1983, now Pat. No. 4,674,830.

(51) **Int. Cl.**

H01S 3/0915	(2006.01)
H01S 3/0933	(2006.01)
H01S 3/094	(2006.01)
H01S 3/067	(2006.01)
H01S 3/06	(2006.01)
H04B 10/17	(2006.01)
H04J 14/02	(2006.01)

(52) **U.S. Cl.** **385/5**; 359/341.3; 359/342; 359/349; 372/6; 372/69

(58) **Field of Classification Search** ..... None

See application file for complete search history.

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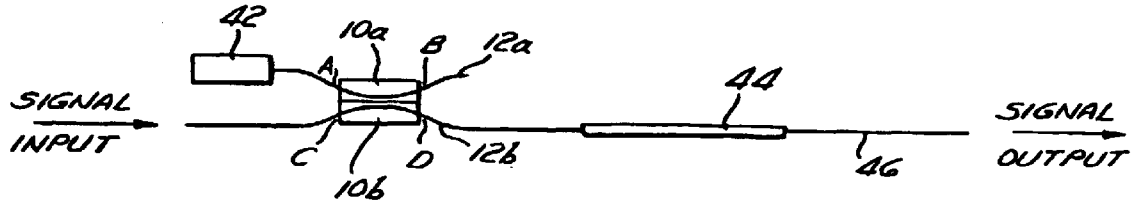
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Primary Examiner—James Menefee

(57) **ABSTRACT**

An amplifier for use with fiber optic systems comprises a neodymium YAG crystal placed in series with a signal-carrying optical fiber. The ND:YAG crystal is supplied by the optical fiber with both the signal to be amplified, and pumping illumination. The pumping illumination is coupled onto the optical fiber by a mutliplexing coupler which is used to combine the signal to be amplified and illumination from a pumping illumination source onto a single optical fiber. The pumping illumination inverts the neodymium ions within the ND:YAG crystal. The signal to be amplified propagates through this crystal to stimulate emission of coherent light from the neodymium ions, resulting in amplification of the signal. Because this arrangement permits the ND:YAG crystal to be end-pumped with pumping illuminations, and because the length of the ND:YAG crystal may be substantially greater than the absorption length for the crystal at the wavelength of the pumping illumination, virtually all of the pumping illumination may be absorbed within the ND:YAG crystal and used for amplification of the signal carried by the optical fiber.



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EX PARTE REEXAMINATION CERTIFICATE ISSUED UNDER 35 U.S.C. 307

THE PATENT IS HEREBY AMENDED AS INDICATED BELOW.

AS A RESULT OF REEXAMINATION, IT HAS BEEN DETERMINED THAT:

The patentability of claims **3**–**4** and **11**–**15** is confirmed.

Claims **1**–**2** and **5**–**6** were previously cancelled.

Claims **7**–**10** and **16**–**22** were not reexamined.

* * * * *